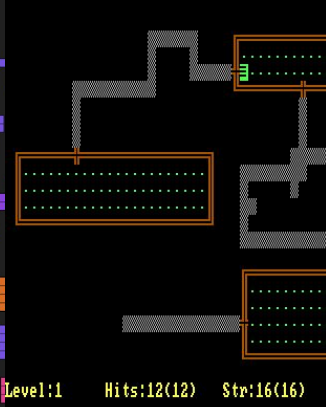




100 100 100
100 100 100
100 100 100

heptominoes (n = 7)



Sol LeWitt, Combinatorial Enumeration, and Rogue

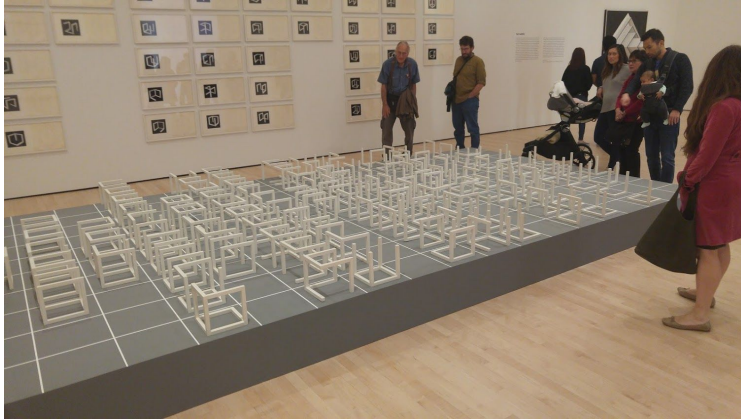
Mark Gritter (novice Artificer)
Roguelike Celebration
October, 2025

Sol LeWitt, "All One, Two, & Three-Part Combinations of Three Transparent Colors", 1985

Labyrinth of Polyominoes, heptominoes

Rogue (PC port)

Sol LeWitt



20th Century
Conceptual Artist,
1928-2007

- wall drawings
- minimalism
- lots of cubes!

← “Incomplete Open
Cubes”, 1974

Photo by the speaker, at San
Francisco Museum of Modern
Art, 2018

Sol LeWitt was a conceptual artist with a broad output during the 20th century. He's maybe most famous for his “wall drawings”, and a lot of his work is geometric.

This photo shows one of my favorite pieces, and during some of the in-person Roguelike Celebrations you could see it up the road at the San Francisco Museum of Modern Art. He created models of all the ways to remove edges from a cube and still have it be connected and self-supporting, or in his words all the ways of “not making a cube”. The art is all 122 of those models!

Wall drawings

Sol LeWitt, "Wall Drawing 289", 1976

A 6-inch (15 cm) grid covering each of the four black walls. White lines to points on the grids. Fourth wall: twenty-four lines from the center, twelve lines from the midpoint of each of the sides, twelve lines from each corner.

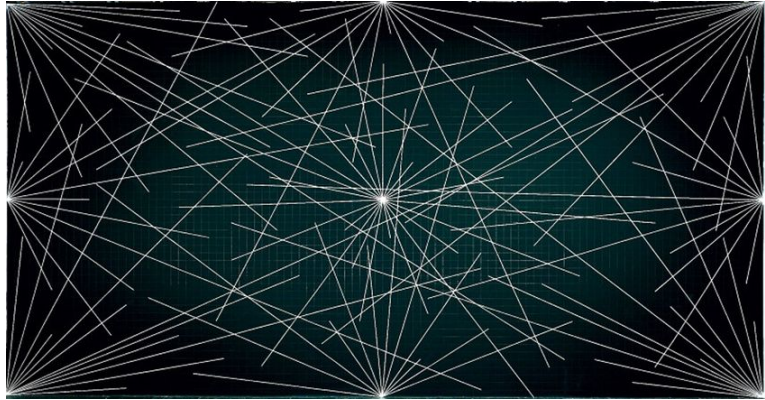


Image from <https://whitney.org/collection/works/25530>

Sol LeWitt's wall drawing series are temporary art, drawn directly on the walls. Each is given as a set of instructions (like those shown here), and often involve exploring all variations of a theme, such as lines in all four orthogonal and diagonal directions, or highlighting all the architectural features of the wall where they are installed.

Wall drawings (2)

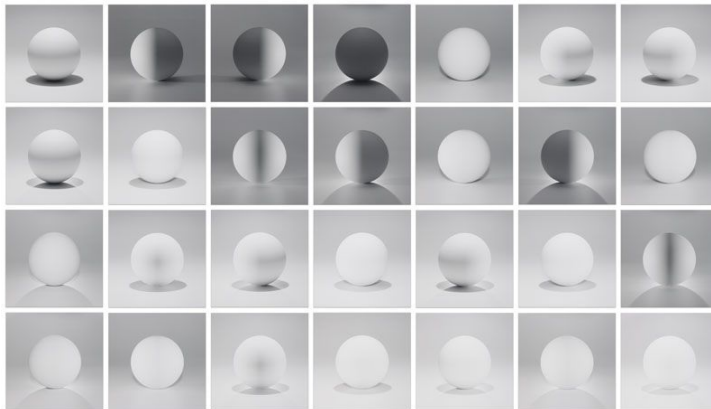
Wall drawings are generative art! They are given as **instructions** rather than as an image. (*)

"The only permanent, concrete form of Wall Drawing #289 is a set of typewritten guidelines and a certificate of authenticity signed by the artist... The exact angle and length of the lines are determined by those who draw them, and the work's precise configuration and scale may be adapted to fit a variety of architectural contexts. Consequently, the wall drawing can differ significantly with each realization." – Whitney Museum of American Art, <https://whitney.org/collection/works/25530>

(*) this separation is ruined a bit by having to hire a draftsman from the LeWitt Studio to execute an "official" drawing.

So I hope the first connection between Sol LeWitt and the Roguelike Celebration is clear – Wall drawings are generative art! They are given as instructions to carry out, rather than an image to be copied. Because of this procedural nature, they can be installed in a variety of spaces.

Serial Art



Sol LeWitt, "A sphere lit from the top, four sides, and all their combinations", 2004

Image from solle Wittprints.org

(P.S. there should be 32, or 31, not 28?)

Many of Sol LeWitt's wall drawings are also exercises in "Serial Art", which explores all the possibilities or consequences of an idea. Here's a more direct example, in which he takes 28 photos of the same sphere, in (almost) all combinations of lighting from the top or four sides.

SKIP: There should be 32, or 31 if we don't include the all-lights-off photo. So LeWitt is not always dogmatic about carrying out the program – it's OK, he's not a mathematician. A mathematician made a similar observation about the incomplete open cubes I showed earlier.

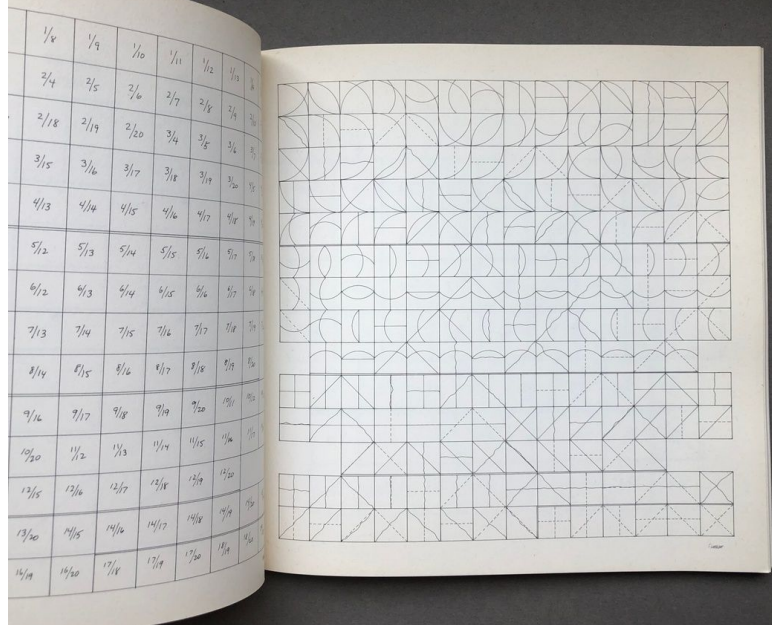
“Arcs and Lines”, Sol LeWitt, 1974

“All combinations of arcs
from four corners, arcs from
four sides, straight lines,
not-straight lines, and
broken lines”

Image from

<https://www.printedmatter.org/catalog/14771/>

(A used copy will set you back
about \$200-\$350 so I don't, alas,
own this.)



Here's a book that LeWitt published with a similar idea – all combinations of arcs from four corners, arcs from four sides, straight lines, not-straight lines, and broken lines.

Serial Art (2)

This takes a further step backwards into abstraction. The artwork is a description of a condition or axiom or law, which is then carried out to produce a tangible result. The artwork typically juxtaposes all of these results together as a finished work.

"The serial artist does not attempt to produce a beautiful or mysterious object but functions merely as a clerk cataloguing the results of his premise." – Sol LeWitt

(I disagree, I think these are beautiful and mysterious. – Mark)

Serial art takes a step further back into abstraction, playing with the idea of eliminating choice or intuition altogether. Serial art starts with a law or set of axioms or a condition, and produces all the figures or artifacts that satisfy that premise. Typically these enumerated combinations are presented together as a single, finished work.

Serial Art (3)

“Three basic operating assumptions separate serially ordered works from multiple variants:

- 1. The derivation of the terms or interior divisions of the work is by means of a numerical or otherwise systematically predetermined process (permutation, progression, rotation, reversal).*
- 2. The order takes precedence over the execution.*
- 3. The completed work is fundamentally parsimonious and systematically self exhausting.”*

– Mel Bochner, “The Serial Attitude”, Artforum, December 1967

(Of course, not everybody agrees with this either – Mark)

The artist Mel Bochner tried to characterize serial art in a similar way:

1. A predetermined process
2. A focus on sequence rather than artistic virtuosity
3. The work is “self-exhausting” – it is done when no more possibilities remain

Coplans' Critique

Coplans defined Serial Art within a context of thematic variation, while also underscoring the importance of repetitive elements or images within an artwork or series. –

<https://www.eykymaclean.com/exhibitions/the-serial-attitude>

*Though often painted in sets, that is, in a limited number that satisfy a given condition, serial images **are nevertheless capable of infinite expansion**. There is no limit to the quantity of works in a series other than what is determined by the artist. Once established, a series may be kept open and added to periodically in the future...*

– John Coplans, “Serial Imagery”, Artforum, October 1986

<https://www.artforum.com/features/with-serial-imagery-the-idea-of-the-masterpiece-is-abandoned-211011/>

“[Bochner] evidently is unaware of the Dedekind-Cantor theory, which is central to the mathematical concept of serial forms.” (!!!)

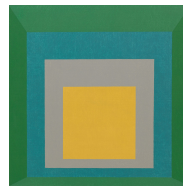
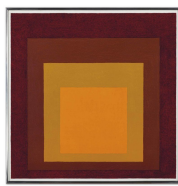
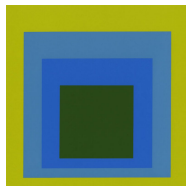
Curator and critic John Coplans (editor of Artforum) disagreed strongly with this definition – he viewed serial art as an act of thematic variation, but one that was open instead of closed. There was no self-exhaustion possible, and artist could always create more instances!

He accused Bochner of ignorance of mathematical set theory. (I think the connection he’s trying to draw is that art is continuous, not discrete, so there is always room for another example.)

Josef Albers, "Homage to the Square". Photo by Selena N. B. H. (cc-by-2.0) at the Tate Modern.



"a rigorously formulaic project comprising more than one hundred paintings and prints and developed over twenty-five years." – MOMA.org



Other 20th-century examples of conceptual show some of the same themes, but not all put together in a way that emphasizes procedurality or exhaustion in the same way that LeWitt does.

Josef Albers' *Homage to the Square* uses a single shape but a variety of colors (many more than are shown here) to study color itself.



Yayoi Kusama,
“Infinity
Mirrored
Room -
Aftermath of
Obliteration
of Eternity”,
2009

from

<https://hirshhorn.si.edu/kusama/infinity-rooms/#aftermath>

One way to think of Yayoi Kusama’s “infinity rooms” as using mirrors to physically embody a rule that causes repetition, and exploring that consequence with mirrored images stretching off into infinity.

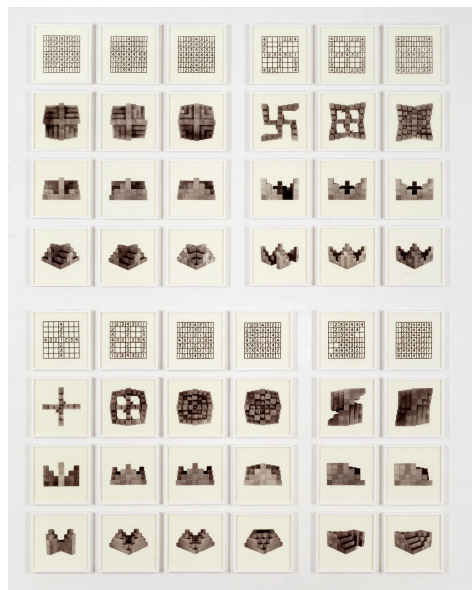
SKIP: My inclusion of Yayoi Kusama is a stretch – in some ways she’s the opposite of a LeWitt, carefully crafting each line and dot. But her mirror rooms, or “infinity rooms” are very much setting a procedural rule in place – one physically embodied by the mirrors – and then exploring the consequences, with mirrored images stretching off into infinity.

Hanne Darboven, (Süd-) Koreanischer Kalender, 1991

<https://spruethmagers.com/artists/hanne-darboven/>



The artist Hanne Darboven often worked with large sequences or grids of numbers; one example is this installation “South-Korean Calendar” with one panel per day from a Korean calendar, annotated with a hand-written equation and some scribbles.



Mel Bochner, "36 Photographs and 12 Diagrams", 1966,

<https://www.glenstone.org/artworks/36-photographs-and-12-diagrams>



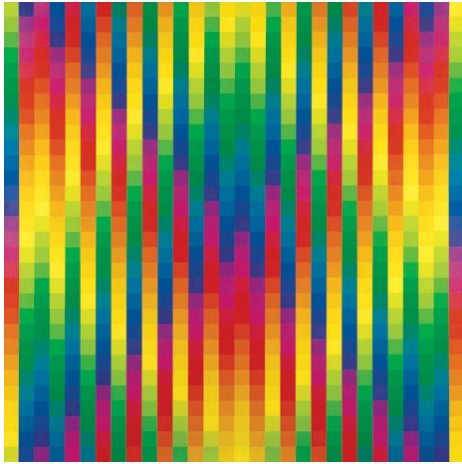
Mel Bochner, "Counting (Double Over)", 1995,

<https://www.melbochnerprints.org/krakow-1994-01/#198>

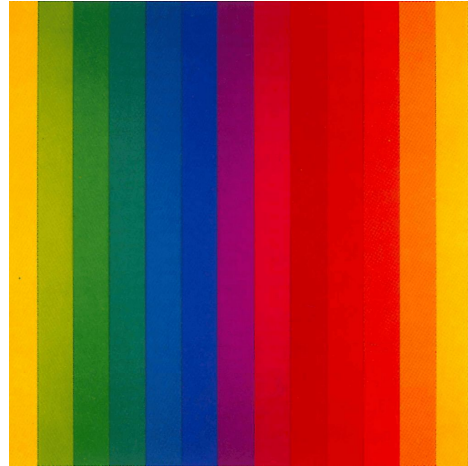
I don't have time to cover as much as I'd like, but here's a quick look at work from Mel Bochner who I quoted earlier.

SKIP: Another example comes from Mel Bochner who I quoted earlier. This piece starts with a diagram describing the sculpture to be constructed out of cubes, which is then carried out and photographed from the same three angles, for nine different diagrams.

Richard Paul Lohse, "Dreissig vertikale systematische
Farbreihen in gelber Rautenform", 1943/1970
from https://www.lohse.ch/works_paintings3_e.html

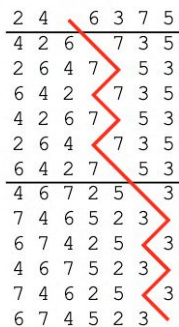


Ellsworth Kelly, "Spectrum IV", 1967
<https://www.miandn.com/exhibitions/ellsworth-kelly/installation-views?view=thumbnails>



and a couple artists using spectrums – a fixed sequence of colors – as their inspiration.

Other Serial Art



2	4	6	3	7	5
4	2	6	7	3	5
2	6	4	7	5	3
6	4	2	7	3	5
4	2	6	7	5	3
2	6	4	7	3	5
6	4	2	7	5	3
4	6	7	2	5	3
7	4	6	5	2	3
6	7	4	2	5	3
4	6	7	5	2	3
7	4	6	2	5	3
6	7	4	5	2	3

“Change Ringing” – playing permutations of a set of bells

- Compositions are called “methods”: each player is given a chart showing how their position in the sequence varies
- $7!$ orders = 5040 changes in a complete method.
- “Composers tend to base their methods in mathematical principles such as group theory” – <https://harpers.org/archive/2025/10/a-change-of-tune-veronique-greenwood-bell-ringing/>
 - You can tell it’s contemporary art because there is active controversy about *what patterns are allowed*

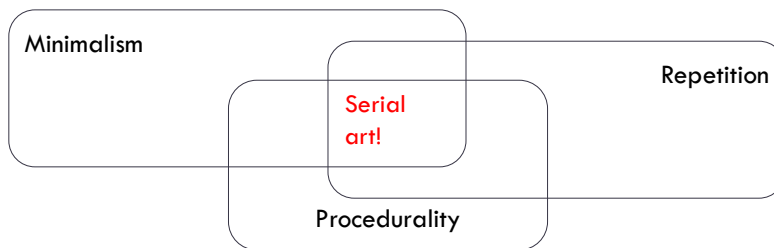
There is an English tradition of playing church bells called “change ringing” which consists of a sequence of permutations of the bells! Here’s a small part of a method showing how bell 1 plays at different times each round. A full change set on 7 bells is a whopping 5040 changes, and people actually do this in real life! It takes hours!

Composers often base their methods in mathematical principles such as group theory, but you can tell it’s contemporary art because there is an active debate about what sort of patterns are allowed in-genre, and which are too physically dangerous or not in keeping with tradition.

Nothing is a clearer genre signifier than debates about what’s in-genre. It happens with roguelikes, it happens with science fiction, with change-ringing, and with the concept of serial art itself. (The quote I showed earlier had a quite scathing rebuttal arguing about the significance or interpretation of sequence!)

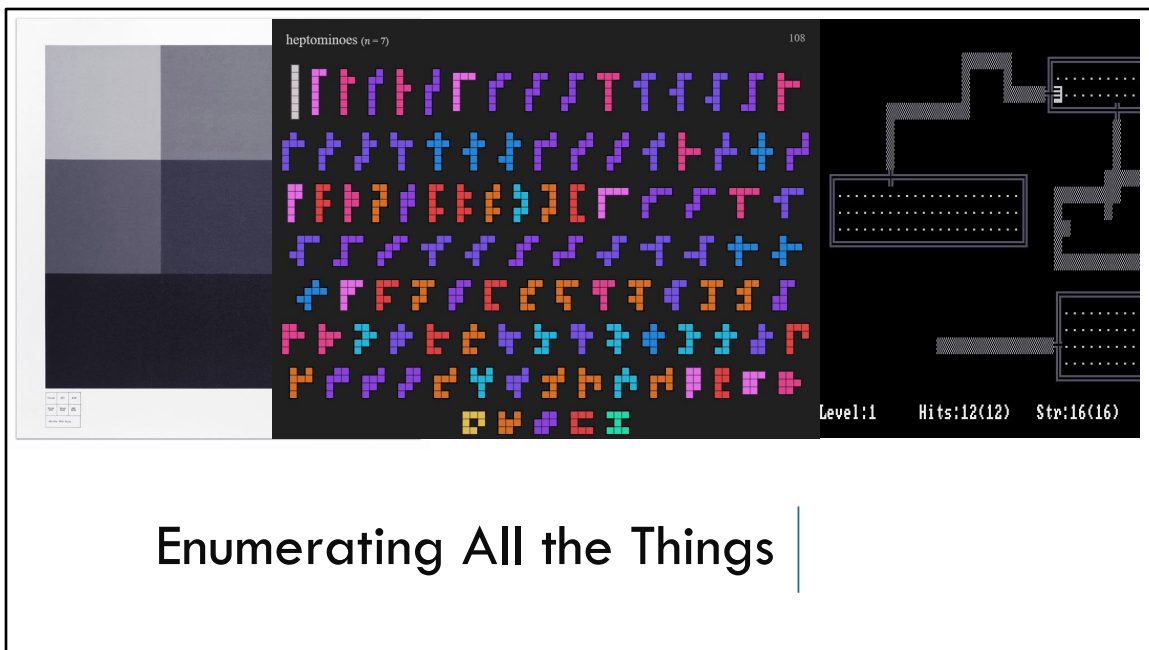
The idea becomes a machine that makes the art.

– Sol LeWitt, “Paragraphs on Conceptual Art”, 1967



This Sol LeWitt quote I think best illustrates the connection between serial art and procedural content generation: “The idea becomes a machine that makes the art.” The artist is not necessarily giving up control, but they are committing themselves to the consequences of a particular idea or algorithm.

Plenty of art is minimalist, or repeats a theme over and over, or procedural, but the sweet spot that I love is the intersection of all three – this is what I view as the core of serial art.



We can find some 21st-century examples of carrying out this process, of enumerating all possibilities, usually using a computer. I think these are serial art too!

The Labyrinth of Polyominoes

By Roguelike
Celebration
2024 speaker
@tesseractis!

All polyominoes
(up to $n=8$) here
arranged by
“genealogy”.

<https://minos.tessera.li/>

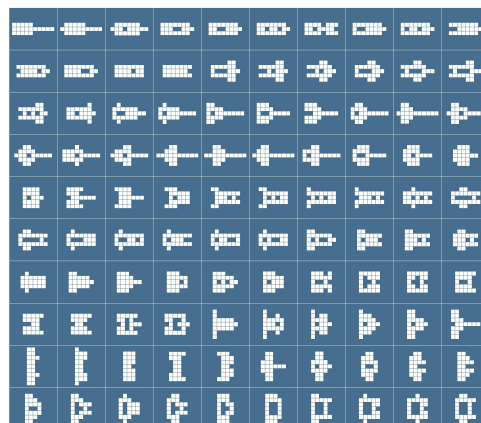


The example I’ve been using in my title slides is Tesseractis’ Labyrinth of Polyominoes, a site showing all possible figures created by sticking squares together (up to 8 squares.) Here’s one view from the site, showing a family tree of polyominoes starting from just a single square. I find this incredibly pleasing and artistic.

All symmetric shapes constructible from the tetrominoes

Jacob Siehler, 2025

<https://mathstodon.xyz/@jsiehler/114744887537037820>

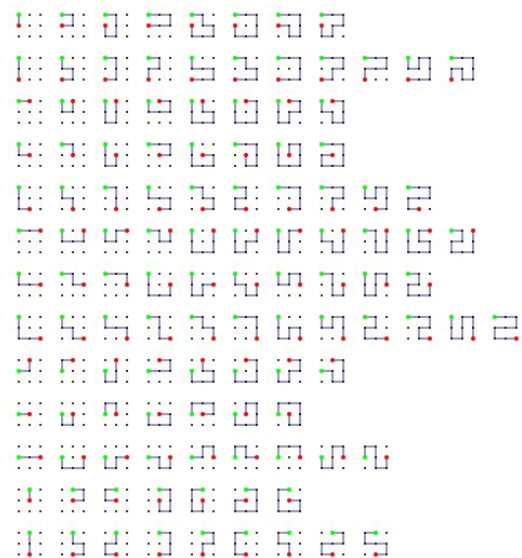


Another polyomino-themed artwork is this enumeration by Jacob Siehler of symmetric shapes constructible using the five tetrominoes.

All simple paths on a 3x3 lattice

Visualization by the speaker

These show directed paths from the green point to the red point, up to symmetries of the rectangle (so, flips but not rotations) and interchange of start and end.



All the Music

Damien Riehl and Noah Rubin,
<https://allthemusic.info/>

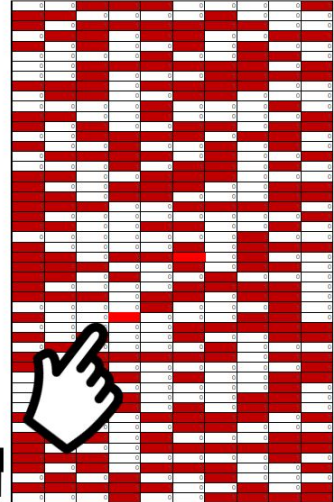
1. Generate every possible melody, up to 10 or 12 notes long.
2. Copyright them all.
3. Stop “substantially similar” music lawsuits forever?

Argument: either Riehl has copyrighted every unused melody, and they are free to use, or melodies aren’t copyrightable.

P.S. in the age of generative AI you should be able to find the unfortunate flaw in this argument.



Finite Melodic Dataset



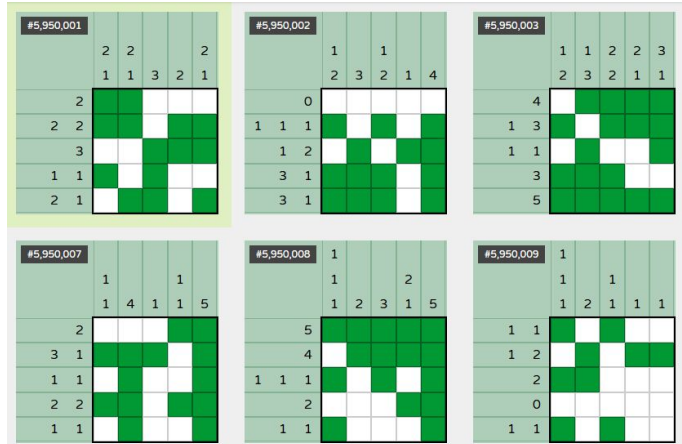
Lawyer Damien Riehl and his collaborator Noah Rubin enumerated all possible melodies – up to 10 notes for the full chromatic scale, or 12 notes for simpler scales. The idea here is to create a tangible artifact of all these melodies, and then copyright them. So this is a political or legal project (and maybe a little bit of trolling) as well as a piece of Serial art.

SKIP: Either all possible melodies are now copyrighted – and Riehl has made them freely available for use – or they were never copyrightable to begin with, and either way lawsuits over melody theft should end.

Every 5x5 Nonogram

Okayest Studio,
<https://pixelogic.app/every-5x5-nonogram>

24,976,511
possibilities, all solved
by humans via a
collaborative effort.



Another online project, exhausting the possible space of 5x5 nonogram puzzles, the ones where you are given a count of the filled-in squares in each row and column of a grid.

There are just shy of 25 million of them, which makes this a pretty big serial art project!

SmallCategories

Ben Spitz,

<https://smallcats.info/>

("in beta")

A "category" is a mathematical object consisting of dots with arrows in between them, obeying a couple simple rules.

164,975 categories each with properties listed, the "multiplication table" of composition, and a visualization.

Quick Reference

Web ID 6ca55ed1-f425-4004-a199-29def9560838

Morphisms 9

Objects 3

Index 59248

Name N/A

Description N/A

Visualization

This feature is still janky! Please excuse the mess.



Morphisms 0 through 2 are identities, shown as objects in this visualization.

Table

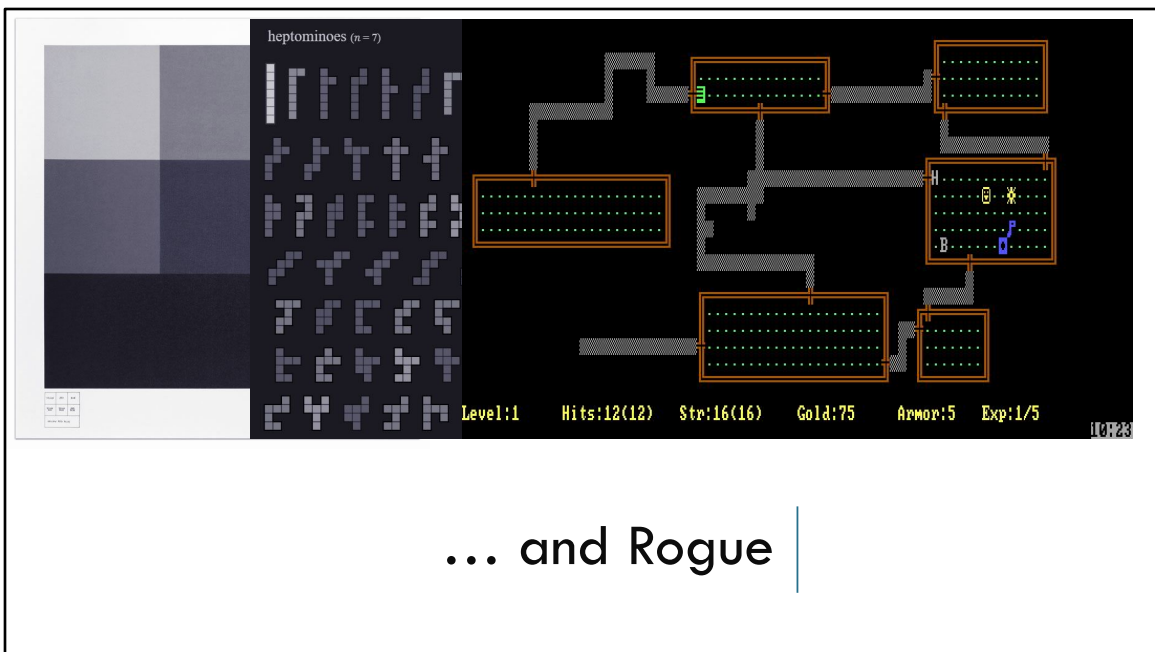
row \ col	0	1	2	3	4	5	6	7	8
0	0	/	/	3	4	5	/	/	/
1	/	1	/	/	/	/	6	7	/
2	/	/	2	/	/	/	/	/	8
3	3	/	/	3	4	5	/	/	/
4	4	/	/	3	4	5	/	/	/
5	5	/	/	3	4	5	/	/	/
6	6	/	/	6	6	6	/	/	/
7	/	7	/	/	/	/	6	7	/
8	/	/	8	/	/	/	/	/	2

Facts

☒ has terminal object	☒ has binary products
☒ is discrete	☒ is terminal
☒ has binary coproducts	☒ is initial object
☒ is grouped	☒ is monoid
☒ is connected	☒ has finite products
☒ is preorder	☒ has equalizers
☒ is skeletal	☒ has finite coproducts
☒ is complete	☒ is cocomplete
☒ has coequalizers	☒ is initial

Dipping back into mathematical objects, Ben Spitz put up a website that lists every small category, as an encyclopedia exhaustively enumerating each possibility.

I think these are all serial art! I love them. They are ideas, that become a machine, that makes the art.



... and Rogue

Which brings us back to Rogue, which I think also fits the bill.

Rogue Level Generator

9 rooms

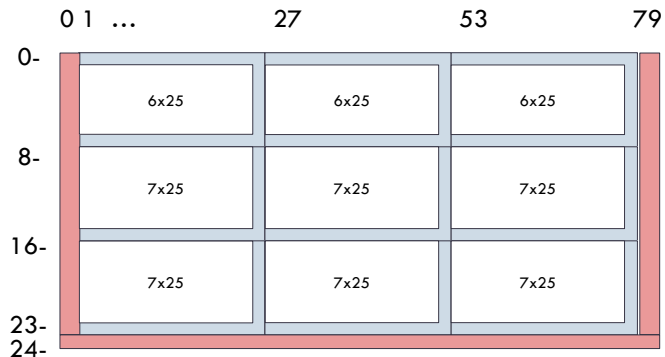
Up to 3 can be missing

For each room:

- Pick a width [4,25]
- Pick a height [4,7]

Pick a location within the “cell” that fits.

Start over if the room would be on the first row (so top rooms only have height 4-6.)



Here's a quick recap of the Rogue level generator. We divide the screen into nine grid elements. Up to three of these rectangles may be left empty. For each rectangle, use a random number generator to pick a room width and a height. Then pick a location for the room so that it fits within the grid rectangle.

SKIP: The first row of rooms is treated a little specially, if we end up putting a room at row 0 we start over with that room's parameters. So the first row is a little shorter, we can't fit a height-7 room there.

Note that there are empty spaces on the left and right sides, and gaps that a room cannot occupy.

Counting the Possibilities

Width 25	Width 24		Width 5	Width 4	
1 horizontal position	2 horizontal positions	...	21 horizontal positions	22 horizontal positions	=253 possibilities

Same calculation for height gives 6 (top row) or 10 (other two rows) possibilities.

So, there are about $(253 \times 6)^3 \times (253 \times 10)^6$ possible combinations of room sizes, if no rooms are absent. That's about $2^{(99.5)}$ possibilities.

- P.S. this number doesn't change much if we account for "gone" rooms

How many level layouts are possible, given this abstract description?

We can count it up and we get 253 possibilities for room width and X-position, and 6 or 10 possibilities for room height and Y-position (the first row has fewer.) This gives nearly 2^{100} possibilities in total, about 10^{30} , just for the rooms themselves – and we haven't considered corridors or gold or monsters.

SKIP: Well, with a maximum-width room we can only put in one horizontal position. With a width 24 room we have two choices. Adding these up $1 + 2 + 3 + \dots + 22$ choices for the minimum width, gives us 253 possibilities. Similarly, for the height we have either 10 or 6 possibilities. This means each of our 3x3 grid of rooms has either 2,530 or 1,518 possibilities. Multiplying those out gives nearly 2^{100} (about 10^{30}) possibilities, just for the rooms themselves – and we haven't considered corridors or gold or monsters.

Rogue as Serial Art

Rogue uses a PRNG (a linear congruential generator) with a 32-bit seed value.

There are only 2^{32} possible versions of a level!

We could enumerate all of them.

- As an art project?
- To better understand the generator?
- I just thought it was cool that not only are there room configurations that are impossible, they *vastly* outnumber the ones that are valid.

But... Rogue uses only a 32-bit seed for its pseudorandom number generator. There are really only 2^{32} possible versions of a given level, about 4 billion. So we could enumerate all of them, as a Serial art project.

Maybe along the way we can understand the generator better. But mainly I just thought it was cool that virtually every room configuration you could write down is actually impossible in Rogue. You have much better odds of playing the lottery than of randomly selecting a valid set of Rogue rooms.

The Rogue RNG(s)

Original rogue:

```
((seed = seed*11109+13849) >> 16) & 0xffff)
```

This is a standard Linear Congruential generator, but using bits 16-32 of the seed avoids some of its worst behaviors.

MS-DOS port:

```
seed *= 125;  
seed -= (seed/2796203) * 2796203;  
return ((ran() + ran())&0x7fffffff1)
```

“This is adapted from the FORTRAN version in ‘Software Manual for the Elementary Functions’ by W.J. Cody, Jr and William Waite.”

A quick aside on the Rogue RNGs – I found something strange, the MS-DOS port replaced the original linear congruential generator with one I frankly find bizarre. The author gave a reference for it, but I don’t trust it. I used the original, but in my code you could swap these out and see what happens.

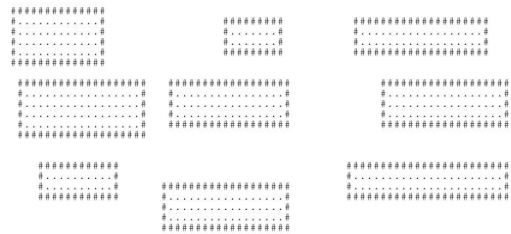
It's not even hard!

I thought that this would take some nontrivial engineering work.

- Parallelize, implement on a GPU, ...?
- Implement an index for reverse lookup ...?

Nope, I copied the implementation into a C++ program (*) and it only takes 11 minutes to run through every possible seed.

(*) = turns out it is not possible to copy Rogue's room generation faithfully without also copying its monster and item generation, oops



first 100 seeds

I initially thought this would be hard – 4 billion is a lot, right? So maybe we'd have to parallelize or build a GPU implementation. But I learned that basically just compiling the code gives a good-enough solution. We can blow through all the possibilities in about 11 minutes of CPU time. This produces an artifact that I think is a piece of Serial Art, even if no human can actually flip through all 4 billion possibilities.

SKIP: But, I have to admit here that I didn't really implement Rogue, because I didn't want to copy all the code to generate monsters and items, which in the original Rogue is interleaved in the level generation. So this is really a study of Rogue-Star, where these steps happened later, like corridor generation does.

Postselection

What if the top-left room is 4x4?

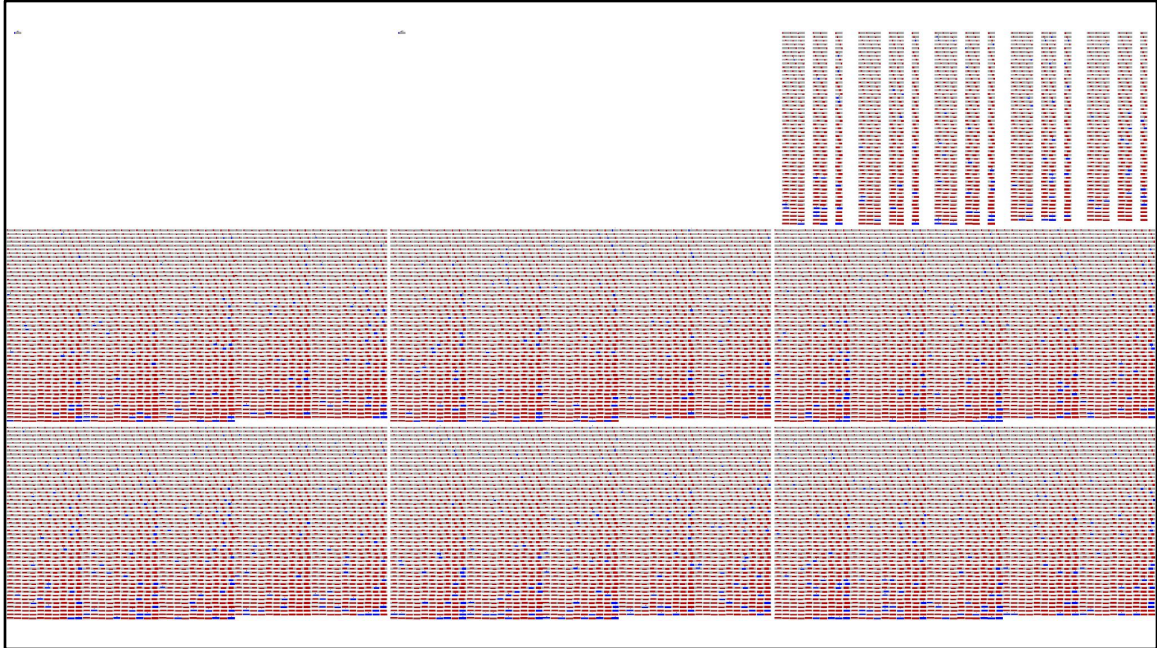
This happens 63,579,000 times. The distribution of the remaining rooms is about even.

(This plot shows how what % of time the given square is occupied.)

[illegible]

But we would like to visualize the result in some way! I don't have 4 billion pixels on my screen.

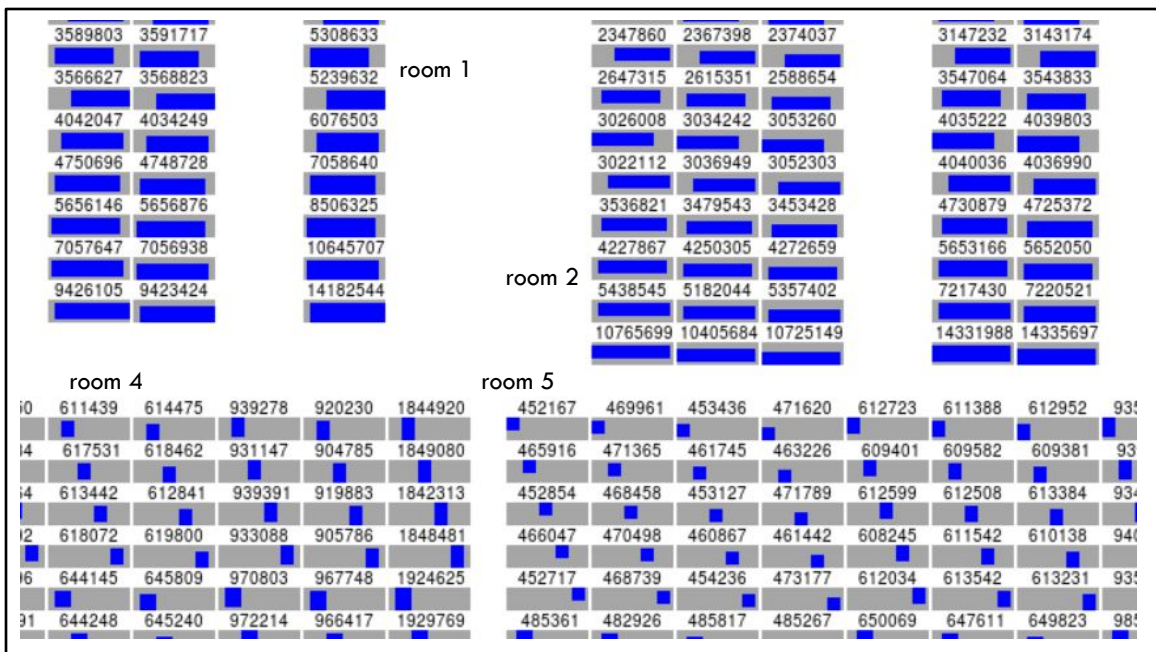
One of the things we can do is look at a subset of our samples created by fixing some feature. For example, what happens if the upper-left room is the smallest possible? Does that affect the composition of other rooms? It shouldn't, and this visualization of how likely a square is to be occupied doesn't show any gross anomalies.



Postselecting on two exact room positions and sizes is enough to turn up impossible values for all the other rooms. If we fix two 4x4 rooms in their upper-left-most positions, that happens for only 209 seeds, not enough to cover all the other choices for the other rooms.

Red = impossible room configurations, blue = possible.

What this means is that we can reconstruct the seed that was used, from as few as three room configurations.



Here's a zoomed in version of the visualization on the previous graph, before preselection – it shows the counts of each room configuration.

Finding extremal seeds

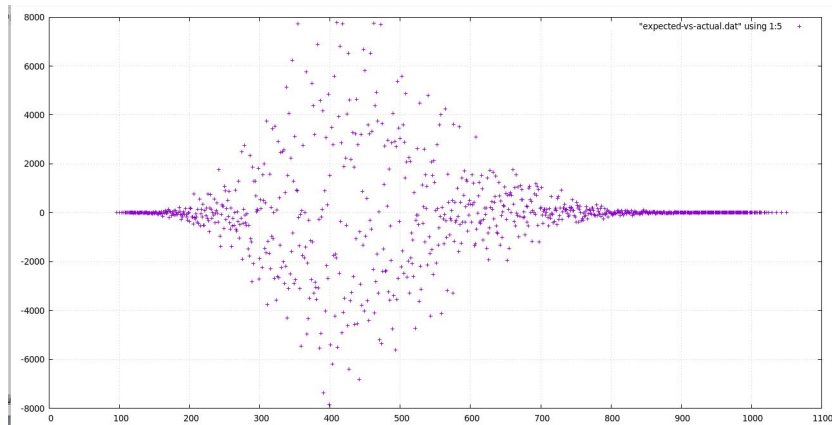
total room area	actual / cumulative	expected cumulative
96-110	0	2
112	2	7
113	4 (6)	9
116	7 (13)	21
117	6 (19)	26
118	5 (24)	30
120	18 (42)	55
121	16 (58)	69
122	11 (69)	82

There seems to be a deficit in how often the smallest areas appear, compared to their expected distribution (in the case where three rooms are gone, so 25% of the sample space.)

Another way to get a handle on this huge collection is to look at extremes – for example, the seeds that produce levels with the smallest or largest total room areas.

A curiosity here is that there seems to be a substantial deficit; there are fewer very small levels than probability would suggest.

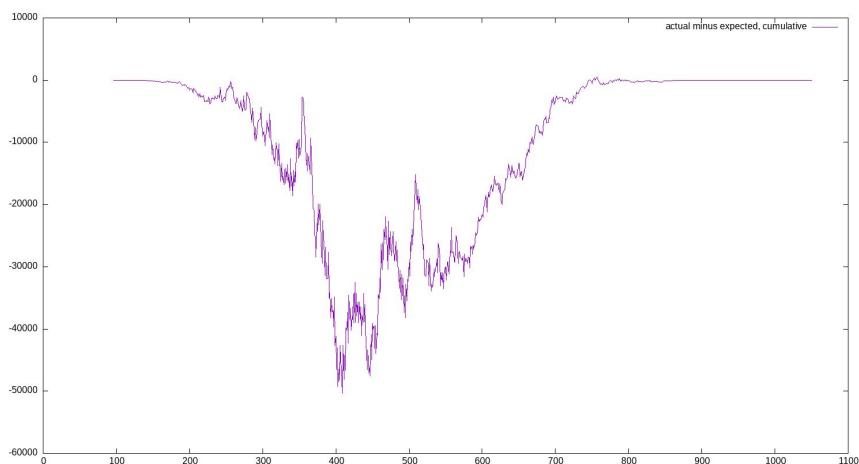
Expected vs. actual (per size)



Here's the plot of variation from expected by size. It has about the shape you'd expect, with more variation in the middle where there are more ways to get a particular area.

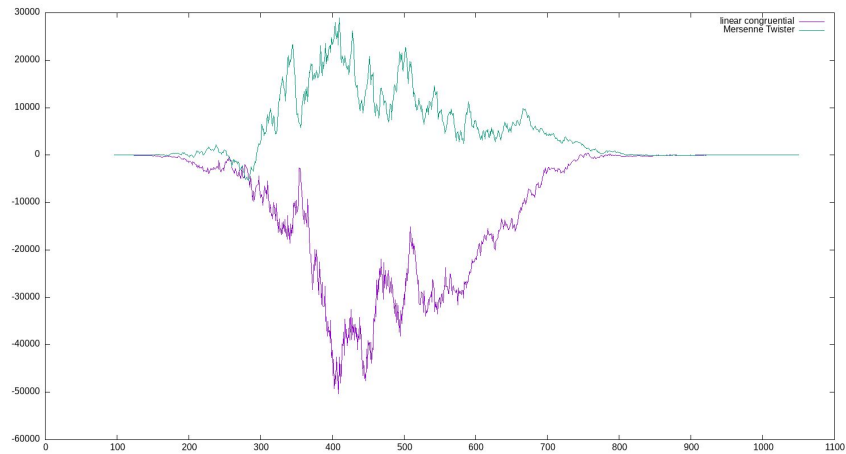
TODO: redo these graphs, use points, proper legend, etc.

Expected vs. actual (cumulative)



Here's a zoomed out version of that same experiment. We see that that negative deficit in small sizes continues for a long, long time. Is this meaningful, or just a random walk? I don't know yet, but my bet is just random.

Expected vs actual (comparison)



Performing the same experiment but with the Mersenne Twister as a PRNG (which runs **much** slower.)

What I'd like to find (but haven't)

- Some artifact of the room distribution that I can point to and say “this is because Rogue uses a linear congruential generator.”
- Some better visualization of the 36-dimensional space of room configurations

So what I'd like to find is some particular artifact I can point to to say “this occurs because we used the linear congruential generator, and disappears if we use a better RNG.”

I'd also like to find better ways to plot these 4 billion data points that come from a 36-dimensional space. My initial attempts at dimensionality reduction didn't turn up anything particularly interesting.

Things to think about

Generators that use 32-bit seeds can be *fully* explored.

- As science!
- As serial art!
- As CYA?!?
- To completely solve a nontrivial roguelike game?

What are other connections between Serial Art and procedural generation?

- I'd love to hear an actual art historian or critic weigh in!

Are there natural examples of games with seeds in the hundreds or millions rather than in the billions?

So what's my point here?

Generators that use 32-bit seeds can be fully explored. It's part science project, part serial art project, and maybe (if you're writing one) you could check that nothing bad ever happens.

I'm hoping that you'll tell me other connections between Serial art and other conceptual art, and procedural generation. I feel like there is a lot more that could be said.

I'm also interested in whether there's some natural middle ground. Are there natural examples of games with hundreds, or no more than millions, of seeds or layouts, so that it would be humanly possible to explore them all?

What other ways can we make ideas, that become machines, that become art?

THANK YOU!



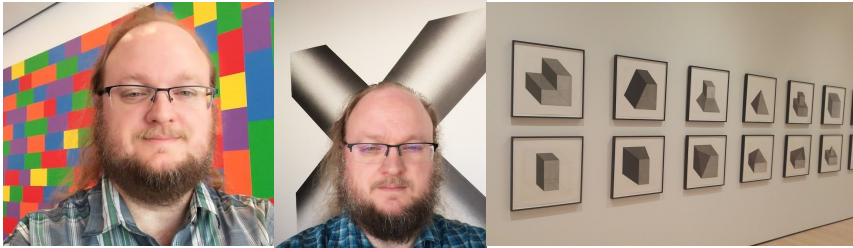
2020 Roguelike
Celebration speaker

On Mastodon: @markgritter@mathstodon.xyz

On Bluesky: @markgritter.bsky.social

On Github: mgritter/rogue-room-generation

In Real Life: Principal Engineer at thirdlaw.io



More Sol LeWitt!
at NYMOMA and
SFMOMA

Thanks for attending, and I'd be happy to interact with you on socials. I've put my materials, including this talk, up on Github.

Things to think about (2)

Building a solver for real roguelike games— could easily “reconstruct” the seed by mistake. :(

- Once we have seen as little as 3 rooms, the value of the initial random seed can be determined, which turns the game from a random one into a deterministic one.

Part of my interest here was looking at building solvers for real Roguelike games. But I realized a solver might end up going down a path where the game has become deterministic because the seed is known, which sort of misses the point.

S3 storage – about \$5/month

Amazon S3 > Buckets > rogue-star-bitmaps

Objects (999+)

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Actions

Create folder

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions [more](#).

Find objects by prefix

< 1 2 3 4 5

☐	Name	Type	Last modified	Size	Storage class
☐	00000000-room0-height4	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-height5	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-height6	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width10	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width11	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width12	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width13	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width14	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width15	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width16	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
☐	00000000-room0-width17	-	July 12, 2025, 15:15:41 (UTC-05:00)	4.0 MB	Standard

Known features of LCGs

- **Marsaglia's theorem:** if used to choose points in an n -dimensional space, the points will lie on at most $(n! \cdot 2^{32})^{1/n}$ hyperplanes
- **No duplicate values** until full cycle completed.
- **Low-order bits have short period;** for example the lowest bit alternates between 0 and 1 each time.

Taking high bits mitigates most of these failures.

Other potential problems

If we compute $\text{rng()} \bmod K$, and K is not a power of 2, then our distribution will be nonuniform:

- 65,536 possible values of the upper 16 bits
- $K=22$ for room width
- Values 0 through 19 have probability $2979/65536$
- Values 20 through 21 have probability $2978/65536$

This is probably too small to have any discernable effect.